

Ethan Wen

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EDUCATION

Princeton University, Princeton, NJ

Expected May 2029

B.S.E. in Electrical & Computer Engineering | Cumulative GPA: 3.82

Fall 2026 Coursework: Contemporary Logic Design (Verilog); Introduction to Programming Systems (C)

PROJECTS

Serial-Controlled VGA Character Display, Verilog, Lattice iCE40-HX1K, Yosys, nextpnr-ice40, IceStorm

- 40x30 character grid at 640x480/60 Hz over UART; 4-state parser FSM, per-pixel font ROM fetch, no framebuffer.
- 585/1280 logic cells, 6/16 block RAMs, closes at 108.5 MHz; per-module benches plus an end-to-end frame compare.

EchoAssist, Swift, SwiftUI, CoreML, Apple Intelligence Foundation Models

Jun 2026 - Jul 2026

- Live speaker-labeled accessibility captions generated fully on-device by two streaming CoreML models (Parakeet ASR, Sortformer diarization) synced on a shared audio clock, running ~0.6s behind real time.
- On-device summarization via Apple Intelligence Foundation Models and translation into six languages.
- Built with a three-person team and submitted to the App Store; review appeal pending.

EXPERIENCE

Software Engineer, Relay

Jun 2026 - Present

- Shipped the friendship, concerts, messaging, and gamification systems for a five-person Flutter music-sharing app.
- Authored engineering design documents covering Firestore data modeling, transaction safety, security rules, and abuse and scale analysis.

HoagieHelp Developer, Hoagie Club

Feb 2026 - Present

- Building frontend components in Next.js, React, TypeScript, and Tailwind for a student academic support platform, with contributions to Python backend services.

iOS App Dev & Machine Learning Program, Everyone Can Code Chicago, Chicago, IL

Jun 2026 - Jul 2026

- Six-week paid cohort program run in partnership between Apple and the City of Chicago; advanced to the second round of the showcase competition, top 12 of 56 teams, judged by Apple engineers.

Digital Creators Intern (Spatial Computing), Project:VISION, Inc., Chicago, IL

Jun 2025 - Jul 2025

- Built AR prototypes in Spline and Reality Composer for Apple Vision Pro; selected to present at the 2025 Everyone Can Code Chicago Showcase.

ACTIVITIES

Princeton University Robotics Club, Princeton, NJ

Sept 2025 - Present

Tinker Tech Lead, Aug 2026 - Present | **Tinker Member**, Sept 2025 - Aug 2026

- Setting project direction and managing the parts budget for Tinker, the club's hardware subteam; firmware, bring-up, and debugging work including vintage iPod refurbishment.

HONORS

Engineering Futures Award, Chicago Engineers' Foundation

\$5,000 merit award, 2026

2nd Place, IEEE SSCS Arduino Contest

2024

SKILLS

Hardware & Digital Design: Verilog; RTL design; Yosys / nextpnr / IceStorm; Icarus Verilog / GTKWave; Lattice iCE40; Arduino / ESP32

Programming: C++; Python; Swift; Dart; TypeScript / JavaScript

Frameworks & Tools: SwiftUI; CoreML; Flutter; Firebase / Firestore; Next.js / React; Git; Xcode; LaTeX

Spoken: English (native); Cantonese (native); Mandarin (professional proficiency)